

# Bounds on the Nonlinearity of Differentially Uniform Functions by Means of Their Image Set Size, and on Their Distance to Affine Functions

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**Abstract**—We revisit and take a closer look at a (not so well known) result of a 2017 paper, showing that the differential uniformity of any vectorial function is bounded from below by an expression depending on the size of its image set. We make explicit the resulting tight lower bound on the image set size of differentially  $\delta$ -uniform functions (which is the only currently known non-trivial lower bound on the image set size of such functions). We also significantly improve an upper bound on the nonlinearity of vectorial functions obtained in the same reference and involving their image set size. We study when the resulting bound is sharper than the covering radius bound. We obtain as a by-product a lower bound on the Hamming distance between differentially  $\delta$ -uniform functions and affine functions, which we improve significantly with a second bound. This leads us to study what can be the maximum Hamming distance between vectorial functions and affine functions. We provide an upper bound which is slightly sharper than a bound by Liu, Mesnager and Chen when  $m < n$ , and a second upper bound, which is much stronger in the case (happening in practice) where  $m$  is near  $n$ ; we study the tightness of this latter bound; this leads to an interesting question on APN functions, which we address (negatively). We finally derive an upper bound on the nonlinearity of vectorial functions by means of their Hamming distance to affine functions and make more precise the bound on the differential uniformity which was the starting point of the paper.

**Index Terms**—Vectorial function, cryptography, substitution box (S-box), nonlinearity, differential uniformity, Hamming distance to affine functions, error correcting code.

## I. INTRODUCTION

**D**IFFERENTIALLY uniform functions are those vectorial functions  $F : \mathbb{F}_2^n \mapsto \mathbb{F}_2^m$  (called  $(n, m)$ -functions) such that the maximum size  $\delta_F$  of the set  $\{x \in \mathbb{F}_2^n; F(x) + F(x + a) = b\}$ , where  $a \in \mathbb{F}_2^n$ ,  $a \neq 0$ , and  $b \in \mathbb{F}_2^m$ , is small. This value  $\delta_F$  is called their differential uniformity. Their study is fundamental for the evaluation of the resistance of the block ciphers which use them as substitution boxes (in brief, S-boxes), against the main attacks (such as the differential attack and the linear attack). S-boxes provide block ciphers with what Claude Shannon [20] called confusion, and are the

only non-linear components of the ciphers; they are therefore essential to their security. Recall that there are two main models of block ciphers: substitution-permutation networks (SPN) and Feistel ciphers; combinations of these two basic models are also possible. In SPN, the cipher is made of rounds being the composition of functions, one of which is the substitution layer, made of the concatenation of S-boxes. Since the cipher must be invertible, the substitution layer needs to be a permutation, and the S-boxes need then to be bijective (with  $m$  necessarily equal to  $n$ ). In Feistel ciphers, the bijectivity of the cipher is ensured by its very structure, and the S-boxes do not need to be bijective ( $m$  can then be different from  $n$ ). It is preferable that the S-boxes are balanced (that is, each element in the co-domain is matched the same number of times as the image of an element of the domain, which needs of course  $m \leq n$ ), but if they are not, then some precautions can be taken like in [19]. The main block cipher used in civil applications, the AES, is an SPN, while the previous one, the DES, was a Feistel cipher. Both models are still important and much used nowadays. The Feistel structure, which is more related to this paper, is used in the block ciphers Blowfish, Camellia, CAST-128, FEAL, GOST, ICE, KASUMI, LOKI97, Lucifer, MARS, MAGENTA, MISTY1, RC5, Simon, TEA, Twofish, XTEA, and even in CLEFIA, MacGuffin, RC2, RC6, Skipjack and SMS4.

Differentially uniform functions also play an important role in coding theory: as shown in [5], the subclass of almost perfect nonlinear functions is directly related to 2-error correcting codes and their duals (which both present much interest for applications to communications and cryptography).

Differentially uniform functions have then been much studied since the 1990's. Recall that the important papers of K. Nyberg, such as [17], [18], have led to the already mentioned AES [11]. But still not enough is known on their properties in general, and since few are known, it is difficult to make conjectures on them. What is known is characterizations by diverse means (see a survey in [4]), but the properties of general differentially uniform functions are essentially unknown (such as their maximum algebraic degree, their minimum and maximum nonlinearities, their minimum and maximum Hamming distances to affine functions, to permutations and to affine permutations, the structure of their image sets, their maximum and minimum numbers of fixed points, etc.). It is essential to understand better the general

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structure and properties of differentially uniform functions, for several reasons. First, the importance of differentially uniform functions for symmetric cryptography and for coding theory automatically calls for a better knowledge of their properties. Second, some features may ease attacks and it is then important to choose the S-boxes so as to minimize this risk. For that, it is necessary to know what are the possible features of general differentially uniform functions. Third, a puzzling question is whether the known differentially uniform functions (which are essentially either power functions over finite fields - possibly slightly modified - or quadratic functions) are peculiar or if on the contrary, they are more or less representative of the general differentially uniform functions. This latter question is wide open, and until new functions are hopefully discovered that would help clarify it, we need to know more about the properties of general differentially uniform functions. But it is impressive how ignorant we are about them. A first step forward has been made in [8] with what we shall recall and develop in Section III, but much more knowledge is needed (this is also true for the subclass of almost perfect nonlinear functions). Ref. [8] is not widely known in the community of vectorial functions for cryptography, since this paper was devoted to side channel attacks, and the bound on the image set size of differentially uniform functions that it contains was not made very explicit. Two preprints have recently presented bounds which are equivalent to it when dealing with characteristic two (the authors of these preprints were not aware that this bound was known):

- Ref. [10] deduces the bound for APN functions from the fact that the sets  $\{x_1 + x_2; x_1, x_2 \in F^{-1}(y), x_1 \neq x_2\}; y \in \mathbb{F}_2^n$ , are pairwise disjoint and have size  $\binom{|F^{-1}(y)|}{2}$ ; then it obtains that  $|Im(F)| \geq \frac{2^n+1}{3}$  if  $n$  is odd and  $|Im(F)| \geq \frac{2^n+2}{3}$  if  $n$  is even; we shall check in Section III that this is also what gives [8] in the particular case of APN functions;

- Ref. [13], which deals with any characteristic and also includes other results, presents a proof rather close to that of [8] (actually, this latter proof generalizes easily to the odd characteristic as we shall observe in Section III).

Of course, the bound is not interesting for cryptography in the framework of substitution permutation networks, since permutations have  $\mathbb{F}_2^n$  for image set, but it is for Feistel ciphers, since the knowledge of the image set and of its size is important for both the designer of a cryptosystem and the attacker.

In the present paper, we first briefly make clear what is proved in [8] on the size of the image sets of differentially uniform functions, since the bound on the image set size by means of the differential uniformity is in fact presented in that paper as a bound on the differential uniformity by means of the image set size. Then, after developing a little more this study, we devote the paper to addressing the important questions of the nonlinearity of differentially uniform functions and their Hamming distance to affine functions. The nonlinearity is an essential parameter of vectorial functions, in the frameworks of cryptography (where it allows to quantify the complexity

of the linear attack, see [14]) and coding theory (where it allows to determine the minimum distance of some linear super-codes of the first order Reed-Muller code, see [16]). The Hamming distance to affine functions plays also a role, as mentioned in [15], since a vectorial function admitting a good approximation by an affine function is cryptographically weak. Also, studying the nonlinearity of general differentially uniform functions being a so hard problem, studying the distance to affine functions may be a way to progress on this subject, since there is a clear relation between these two values (this relation needs to be quantified precisely, though; we know already that the nonlinearity is bounded above by this Hamming distance and we give a complementary bound in Corollary 2). It is puzzling that the maximum distance from vectorial functions to affine functions is unknown, and it would be very interesting to know what are the vectorial functions which lie at maximum distance to affine functions (i.e. the functions playing the role of bent vectorial functions with respect to this other nonlinearity parameter that is the distance to affine functions); maybe these functions have some interesting properties.

When  $\delta_F = 2$  (which is optimum), differentially  $\delta$ -uniform  $(n, n)$ -functions are called APN (almost perfect nonlinear). We shall of course be particularly interested in these functions, since they contribute in an optimal way to the resistance against the differential attack.

The paper is organized as follows.

After preliminaries in Section II, we revisit and study more in detail in Section III the result from [8] on the differential uniformity of vectorial functions, given the size of their image sets. We study the equivalent tight lower bound on the image set size of differentially  $\delta$ -uniform functions. We apply this lower bound to the sums of  $F$  and affine functions in Section IV and pose the natural question whether the resulting property of APN functions allows to characterize them; we answer negatively. We observe in Section V that an upper bound given in [8] on the nonlinearity of  $(n, m)$ -functions by means of their image set size is weak, and we derive a much better bound. We study when this bound improves upon the covering radius bound. In Section VI, we also bound from below the Hamming distance between differentially uniform functions and affine functions, first as a consequence of the bound on the image set size and then by an improved bound. This shows that the resistance against differential attacks ensures automatically the resistance against attacks by affine approximations. This leads us in Section VII to study the maximum Hamming distance between vectorial functions and affine functions and to first slightly improve upon the only known explicit upper bound on it and second significantly improve upon it when  $m$  is near  $n$ . Showing that this latter bound is not tight leads to an interesting question on APN functions that we solve. In Section VIII, we derive an upper bound on the nonlinearity of any  $(n, m)$ -function  $F$  by an expression depending on its Hamming distance to affine functions. In Section IX, we make the bound on the differential uniformity of  $F$  more accurate, by introducing another parameter of  $F$ .

## II. PRELIMINARIES

We shall denote by  $0_n$  (resp.  $1_n$ ) the zero vector (resp. the all-1 vector) of length  $n$  and by  $e_i$  the  $i$ -th vector of the canonical basis of the vector space  $\mathbb{F}_2^n$ . We denote by  $w_H(x)$  the Hamming weight of an element  $x$  of  $\mathbb{F}_2^n$ , that is, the size of its support  $\text{supp}(x) = \{i \in \{1, \dots, n\}; x_i = 1\}$ . We call co-support of  $x$  the complement of its support.

The vector space  $\mathbb{F}_2^n$  is sometimes endowed with the structure of the field  $\mathbb{F}_{2^n}$  (with zero element denoted by 0); indeed, this field being an  $n$ -dimensional vector space over  $\mathbb{F}_2$ , each of its elements can be identified with the binary vector of length  $n$  of its coordinates relative to a fixed basis.

Given an  $n$ -variable Boolean function  $f$ , that is, a function from  $\mathbb{F}_2^n$  to  $\mathbb{F}_2$ , the Walsh transform of  $f$  is defined as  $W_f(u) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + u \cdot x}$ , where “ $\cdot$ ” is some chosen inner product in  $\mathbb{F}_2^n$  (such as  $u \cdot x = \sum_{i=1}^n u_i x_i$ , or, if  $\mathbb{F}_2^n$  is endowed with the structure of  $\mathbb{F}_{2^n}$ ,  $u \cdot x = \text{tr}_n(ux)$ , where  $\text{tr}_n(x) = \sum_{i=0}^{n-1} x^{2^i}$  is the so-called absolute trace function). The Walsh transform satisfies the *inverse Walsh transform relation*:

$$\sum_{u \in \mathbb{F}_2^n} W_f(u) (-1)^{u \cdot v} = 2^n (-1)^{f(v)}, \forall v \in \mathbb{F}_2^n. \quad (1)$$

For a given  $(n, m)$ -function  $F$ , that is, a function from  $\mathbb{F}_2^n$  to  $\mathbb{F}_2^m$ , the value  $W_F(u, v)$  of the Walsh transform of  $F$  at  $(u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2^m$  equals by definition that of the Walsh transform of the Boolean function  $v \cdot F$  at  $u$ :

$$W_F(u, v) = \sum_{x \in \mathbb{F}_2^n} (-1)^{v \cdot F(x) + u \cdot x},$$

where we use the same notation “ $\cdot$ ” for inner products in  $\mathbb{F}_2^n$  and  $\mathbb{F}_2^m$ . The *nonlinearity* of a Boolean function  $f$  equals its minimum Hamming distance to affine Boolean functions, that is, to functions of the form  $u \cdot x + \epsilon$ ,  $u \in \mathbb{F}_2^n$ ,  $\epsilon \in \mathbb{F}_2$ . It equals then:

$$nl(f) = 2^{n-1} - \frac{1}{2} \max_{u \in \mathbb{F}_2^n} |W_f(u)|. \quad (2)$$

It is bounded above by  $2^{n-1} - 2^{\frac{n}{2}-1}$ , according to the covering radius bound (see e.g. [4]) and  $f$  is called *bent* if it achieves this value. The nonlinearity of an  $(n, m)$ -function  $F$  equals the minimum nonlinearity of its component functions  $v \cdot F$ ,  $v \in \mathbb{F}_2^m \setminus \{0_m\}$ . It equals then

$$nl(F) = 2^{n-1} - \frac{1}{2} \max_{\substack{u \in \mathbb{F}_2^n \\ v \in \mathbb{F}_2^m, v \neq 0_m}} |W_F(u, v)|. \quad (3)$$

It quantifies the contribution of the  $(n, m)$ -function, when used as an S-box, to the resistance of block ciphers against the linear attack [14]. It is bounded above by  $2^{n-1} - 2^{\frac{n}{2}-1}$  as well and  $F$  is called *bent* if it achieves this value. Bent functions exist for  $m \leq \frac{n}{2}$ ,  $n$  even, only [17]. For  $m = n$ ,  $nl(F)$  is bounded above by  $2^{n-1} - 2^{\frac{n-1}{2}}$ , according to the Sidelnikov-Chabaud-Vaudenay (SCV) bound [9] and  $F$  is called *almost bent* (AB) if it achieves this value (AB functions exist for every odd  $n$ , see e.g. [4] for a description of known ones).

Any  $(n, m)$ -function can be uniquely represented by its algebraic normal form (ANF):

$$F(x) = \sum_{I \subseteq \{1, \dots, n\}} a_I \left( \prod_{i \in I} x_i \right) = \sum_{I \subseteq \{1, \dots, n\}} a_I x^I, \quad (4)$$

where  $a_I$  belongs to  $\mathbb{F}_2^m$ . The global degree of the ANF is called the algebraic degree of  $F$ . It equals the maximum algebraic degree of the component functions of  $F$ . The zero function has by convention algebraic degree 0. Affine functions are those functions of algebraic degree at most 1. If  $\mathbb{F}_2^n$  is endowed with the structure of the field  $\mathbb{F}_{2^n}$ , then every  $(n, n)$ -function (and then, every  $(n, m)$ -function where  $m$  divides  $n$ ) can be uniquely represented by its univariate representation:

$$F(x) = \sum_{i=0}^{2^n-1} \delta_i x^i \in \mathbb{F}_{2^n}[x]/(x^{2^n} + x). \quad (5)$$

Hence, every  $(n, n)$ -function is a polynomial function. The algebraic degree equals then  $\max_{j=0, \dots, 2^n-1; \delta_j \neq 0} w_2(j)$ , where  $w_2$  denotes the Hamming weight of the binary expansion. The functions whose univariate expression is a monomial are called *power functions*.

We shall denote the image set  $\{F(x); x \in \mathbb{F}_2^n\}$  of  $F$  by  $\text{Im}(F)$ .

An  $(n, m)$ -function  $F$  is called differentially  $\delta$ -uniform, for a given positive integer  $\delta$ , if for every  $a \in \mathbb{F}_2^n \setminus \{0_n\}$  and every  $b \in \mathbb{F}_2^m$ , the equation  $F(x) + F(x + a) = b$  has at most  $\delta$  solutions. We denote the minimum of these integers  $\delta$  by  $\delta_F$  and call it the differential uniformity of  $F$ . It quantifies the contribution of the  $(n, n)$ -function, when used as an S-box, to the resistance of block ciphers against the differential attack [1]. For every  $(n, m)$ -function  $F$ , we have  $\delta_F \geq \max(2, 2^{n-m})$ . It is shown in [17] that, for  $m < n$ , equality is equivalent to the fact that  $F$  is bent, and this can then happen if and only if  $n$  is even and  $m \leq \frac{n}{2}$ .

Note that we can have  $\delta_F = 2$  only when  $n \geq m$ . An  $(n, n)$ -function  $F$  is called *almost perfect nonlinear* (APN) if it is differentially 2-uniform, that is, if for every  $a \in \mathbb{F}_2^n \setminus \{0_n\}$  and every  $b \in \mathbb{F}_2^n$ , the equation  $F(x) + F(x + a) = b$  has 0 or 2 solutions (i.e. the derivative  $D_a F(x) = F(x) + F(x + a)$  is 2-to-1). Equivalently,  $|\{D_a F(x), x \in \mathbb{F}_2^n\}| = 2^{n-1}$ . Still equivalently, for distinct elements  $x, y, z, t$  of  $\mathbb{F}_2^n$ , the equality  $x + y + z + t = 0_n$  implies  $F(x) + F(y) + F(z) + F(t) \neq 0_n$ , that is, the restriction of  $F$  to any 2-dimensional flat (i.e. affine plane) of  $\mathbb{F}_2^n$  is non-affine, that is, for every linearly independent  $a, b \in \mathbb{F}_2^n$ , the function  $D_a D_b F(x)$  does not vanish. There are several characterizations of APN functions (see e.g. the survey [4]). The relationship between nonlinearity and differential uniformity is not clear for APN functions: all that is known on the nonlinearity of general APN  $(n, n)$ -functions is that it cannot be 0 (see [2]), while all known APN functions have a rather good nonlinearity. Note that differentially uniform functions can have nonlinearity 0 (for instance, an  $(n, n)$ -function obtained from an APN  $(n, n)$ -function by replacing one of its coordinate functions by an affine Boolean function is differentially 4-uniform and has nonlinearity 0).



### III. THE LOWER BOUND ON THE SIZE OF THE IMAGE SETS OF DIFFERENTIALLY UNIFORM FUNCTIONS

In [8, Subsection 4.2] is studied (for reasons related to side channel attacks that we shall not develop here) the differential uniformity of those  $(n, n)$ -functions  $F$  satisfying, for some  $d$ , that  $d_H(x, F(x)) \leq d$  for every  $x \in \mathbb{F}_2^n$ . The differential uniformity of such functions is shown to be bad if  $d$  is too small. The authors observe that the condition being that all the images of the function  $F(x) + x$  have Hamming weight at most  $d$ , the size of the image set of this latter function (which has the same differential uniformity as  $F$ ) is then bounded above by  $D = \sum_{i=0}^d \binom{n}{i}$ . A lower bound is then proved on the differential uniformity of a function by means of the size of its image set.

We now want to study more deeply the incidence of the image set size of differentially uniform  $(n, m)$ -functions, since it is only approached as a tool in this paper, whereas it deserves more attention because the bound of [8] is one of the rare known properties of differentially uniform functions. For our present paper to be self-contained, we briefly recall what is the lower bound in [8] and the method for proving it. Let  $F$  be any  $(n, m)$ -function. We have  $\sum_{a \in \mathbb{F}_2^n; a \neq 0_n} |(D_a F)^{-1}(0_m)| = |\{(x, y) \in (\mathbb{F}_2^n)^2; F(x) = F(y)\}| - 2^n = \sum_{b \in \text{Im}(F)} |F^{-1}(b)|^2 - 2^n$ .

$2^n \geq \frac{(\sum_{b \in \text{Im}(F)} |F^{-1}(b)|)^2}{|\text{Im}(F)|} - 2^n = \frac{2^{2n}}{|\text{Im}(F)|} - 2^n$ , where the inequality is obtained by the Cauchy-Schwarz inequality. Since, in every numerical sequence, there exists an element larger than or equal to the arithmetic mean of the sequence, we deduce that there exists  $a \in \mathbb{F}_2^n$ , nonzero, such that  $|D_a F^{-1}(0_m)| \geq \frac{2^{2n}}{|\text{Im}(F)|} - 2^n$ . We have then:

*Proposition 1:* [8] For every  $(n, m)$ -function, the differential uniformity of  $F$  satisfies:

$$\delta_F \geq \left\lceil \frac{\frac{2^{2n}}{|\text{Im}(F)|} - 2^n}{2^n - 1} \right\rceil.$$

Equivalently, we have  $\frac{2^{2n}}{|\text{Im}(F)|} \leq (2^n - 1)\delta_F + 2^n$ , that is:

$$|\text{Im}(F)| \geq \left\lceil \frac{2^{2n}}{(2^n - 1)\delta_F + 2^n} \right\rceil \geq \left\lceil \frac{2^n}{\delta_F + 1} \right\rceil. \quad (6)$$

For  $\delta_F = 2$ , we have then that for every APN  $(n, n)$ -function:

$$|\text{Im}(F)| \geq \left\lceil \frac{2^{2n}}{3 \cdot 2^n - 2} \right\rceil. \quad (7)$$

Note that  $\left\lceil \frac{2^{2n}}{3 \cdot 2^n - 2} \right\rceil$  equals  $\frac{2^n + 1}{3}$  when  $n$  is odd and  $\frac{2^n + 2}{3}$  when  $n$  is even; these exact values are given in [10].

We observe that the bound (7) is much stronger than the bound  $|\text{Im}(F)| \geq \frac{1 + \sqrt{2^{2n} - 7}}{2}$  obtained in [21]. Note also that it is tight. Indeed, we know that APN power functions in even dimension  $n$  have for image set the set of cubes in  $\mathbb{F}_{2^n}$  (see e.g. [4]), whose number equals  $1 + \frac{2^n - 1}{3} = \frac{2^n + 2}{3}$ , which achieves then the bound of (7) with equality. This is natural, given the proof above of the bound, since, for every power

APN function  $F$ , the size of  $F^{-1}(b)$  is independent of  $b \neq 0_n$  in  $\text{Im}(F)$ , and the sequence  $|F^{-1}(b)|$ ,  $b \in \text{Im}(F)$ , is then constant except at  $0_n$ , and the Cauchy-Schwarz inequality is close to an equality.

*Remark:* Proposition 1 generalizes straightforwardly to any characteristic: let  $p$  be a prime and  $F : \mathbb{F}_p^n \mapsto \mathbb{F}_p^m$ . Denoting  $D_a F(x) = F(x + a) - F(x)$ , we have  $\sum_{a \in \mathbb{F}_p^n; a \neq 0_n} |(D_a F)^{-1}(0_m)| = |\{(x, y) \in (\mathbb{F}_p^n)^2; F(x) = F(y)\}| - p^n = \sum_{b \in \text{Im}(F)} |F^{-1}(b)|^2 - p^n \geq \frac{(\sum_{b \in \text{Im}(F)} |F^{-1}(b)|)^2}{|\text{Im}(F)|} - p^n = \frac{p^{2n}}{|\text{Im}(F)|} - p^n$ , and there exists  $a \in \mathbb{F}_p^n$ , nonzero, such that  $|D_a F^{-1}(0_m)| \geq \frac{p^{2n}}{|\text{Im}(F)|} - p^n$ . We have then:  $\delta_F \geq \left\lceil \frac{p^{2n}}{|\text{Im}(F)|} - p^n \right\rceil$ . This is equivalent to:  $|\text{Im}(F)| \geq \left\lceil \frac{p^{2n}}{(p^n - 1)\delta_F + p^n} \right\rceil \geq \left\lceil \frac{p^n}{\delta_F + 1} \right\rceil$ .  $\diamond$

### IV. ON THE SUMS OF DIFFERENTIALLY UNIFORM FUNCTIONS AND AFFINE FUNCTIONS

The bound of Proposition 1 applies to  $F + A$ , where  $A$  is any affine function (or equivalently, to  $F + L$ , where  $L$  is any linear function). The next corollary is then straightforward but it gives an interesting property, which may for instance eliminate a large number of potential APN candidates.

*Corollary 1:* Let  $F$  be any differentially  $\delta$ -uniform  $(n, m)$ -function. Let  $\mathcal{A}$  be the set of affine  $(n, m)$ -functions<sup>1</sup>. Then, for every  $A \in \mathcal{A}$ , we have:

$$|\text{Im}(F + A)| \geq \left\lceil \frac{2^{2n}}{(2^n - 1)\delta + 2^n} \right\rceil.$$

In particular, an  $(n, n)$ -function can be APN only if, for every  $A \in \mathcal{A}$ , we have:

$$|\text{Im}(F + A)| \geq \left\lceil \frac{2^{2n}}{3 \cdot 2^n - 2} \right\rceil. \quad (8)$$

Hence, when searching for APN functions, we can eliminate as potential candidates, for each  $A$ , all the functions  $F$  such that  $|\text{Im}(F + A)| < \left\lceil \frac{2^{2n}}{3 \cdot 2^n - 2} \right\rceil$ .

Trying to build vectorial functions satisfying (8) for every affine function  $A$  (that is, for every linear function) without using the notion of APN function and Corollary 1 seems hard. Even for the simplest examples that are Gold and inverse functions, it seems difficult to prove directly that for every linear function  $L$ , the sizes of the sets  $\{x^{2^k+1} + L(x); x \in \mathbb{F}_{2^n}\}$  for  $k$  co-prime with  $n$  and  $\{x^{2^n-2} + L(x); x \in \mathbb{F}_{2^n}\}$  for  $n$  odd are larger than or equal to  $\frac{2^{2n}}{3 \cdot 2^n - 2}$ .

*Remark:* Let us try to see if we have the same with  $x^{2^k+1}$  for  $n$  not co-prime with  $k$ . Note that taking for  $L$  the zero function, the size of  $\text{Im}(F)$  is equal to  $1 + \frac{2^n - 1}{\gcd(2^n - 1, 2^k + 1)} = 1 + \frac{(2^n - 1)\gcd(2^n - 1, 2^k - 1)}{\gcd(2^n - 1, 2^{2k} - 1)} = 1 + \frac{(2^n - 1)(2^{\gcd(n, k)} - 1)}{(2^{\gcd(n, 2k)} - 1)} =$

$$\begin{cases} 2^n & \text{if } \text{val}_2(k) \geq \text{val}_2(n) \\ 1 + \frac{2^n - 1}{2^{\gcd(n, k)} + 1} & \text{if } \text{val}_2(k) < \text{val}_2(n) \end{cases},$$

<sup>1</sup>We use the same symbol as for affine Boolean functions since there is no ambiguity.

where  $\text{val}_2(k)$  is the 2-valuation of  $k$ . Hence, if  $\text{val}_2(k) < \text{val}_2(n)$ , we have  $|Im(F)| < \frac{2^{2n}}{3 \cdot 2^{n-2}}$ , since  $2^{\text{gcd}(n,k)} + 1 \geq 5$  (because  $n$  is assumed not co-prime with  $k$ ) and  $1 + \frac{2^{n-1}}{5} = \frac{2^n + 4}{5} < \frac{2^{2n}}{3 \cdot 2^{n-2}}$ , for  $n \geq 3$ . In the case  $\text{val}_2(k) \geq \text{val}_2(n)$ , we would need to consider nonzero  $L$  and the case seems more complex.

Similarly, it seems difficult to say if we have the same with  $x^{2^n-2}$  for  $n$  even as for  $n$  odd. For  $L = 0$ , the inequality is satisfied since the inverse function is a permutation and for all the other affine functions  $L$  we already know that  $|Im(F)| \geq \left\lceil \frac{2^{2n}}{5 \cdot 2^{n-4}} \right\rceil$ , since  $F + L$  is differentially 4-uniform, but it seems difficult to say more. For  $L(x) = x^{2^k}$ , the equation  $x^{2^n-2} + L(x) = b$  for  $b \neq 0$  is equivalent to  $x^{2^k+1} + bx = 1$  and by the change of variable  $x \rightarrow b^{2^{-k}}x$ , to  $x^{2^k+1} + x = b^{-(1+2^{-k})}$ . This equation can be handled as shown in [12], [22], but it is already quite complex; addressing all other linear functions  $L$  seems out of reach. It is even difficult to know what can be the largest possible value of  $\sum_{a \in \mathbb{F}_2^n; a \neq 0} |(D_a F + L)^{-1}(0)| = \sum_{a \in \mathbb{F}_2^n; a \neq 0} |(D_a F)^{-1}(L(a))|$  (which provides the lower bound on  $|Im(F+L)|$ ). We know that  $|(D_a F)^{-1}(b)|$  equals 4 if and only if  $ab = 1$  and equals 2 if and only if  $ab \notin \mathbb{F}_2$  and  $\text{tr}_n(\frac{1}{ab}) = 0$ . It is difficult to say if there exist linear functions such that  $|(D_a F)^{-1}(L(a))| \geq 2$  for every nonzero  $a$ , and how many times  $|(D_a F)^{-1}(L(a))|$  can then reach 4.

In the case of Gold functions, it may be easier to determine the largest possible value of  $\sum_{a \in \mathbb{F}_2^n; a \neq 0} |(D_a F)^{-1}(L(a))| = \sum_{a \in \mathbb{F}_2^n; a \neq 0} |\{x \in \mathbb{F}_2^n; ax^{2^k} + a^{2^k}x = L(a) + a^{2^k+1}\}| = \sum_{a \in \mathbb{F}_2^n; a \neq 0} |\{x \in \mathbb{F}_2^n; x^{2^k} + x = \frac{L(a)}{a^{2^k+1}} + 1\}|$ .

Proving directly (without using Corollary 1) that, over  $\mathbb{F}_{2^n}$ , Kasami functions  $x^{4^i-2^i+1}$ ,  $\text{gcd}(i, n) = 1$ , Welch functions  $x^{2^{\frac{n-1}{2}+3}}$ ,  $n$  odd, Niho functions  $x^{2^{(n-1)/2+2(n-1)/4-1}}$  if  $n \equiv 1 \pmod{4}$ , and  $x^{2^{(n-1)/2+2(3n-1)/4-1}}$  if  $n \equiv 3 \pmod{4}$ , and Dobbertin functions  $x^{2^{\frac{4n}{5}+2\frac{3n}{5}+2\frac{2n}{5}+2\frac{n}{5}-1}}$ ,  $5|n$ , satisfy that, for every affine  $(n, n)$ -function  $L$ , we have  $|Im(F+L)| \geq \left\lceil \frac{2^{2n}}{3 \cdot 2^{n-2}} \right\rceil$ , seems still more difficult than for Gold and inverse APN functions.  $\diamond$

Corollary 1 leads to the natural question whether its converse is true: given an  $(n, n)$ -function  $F$ , if for every affine (or every linear)  $(n, n)$ -function  $L$ , we have  $|Im(F+L)| \geq \left\lceil \frac{2^{2n}}{3 \cdot 2^{n-2}} \right\rceil$ , then  $F$  is it necessarily APN?

The answer to this question is no: there are already counter-examples with functions in dimension 4. For instance, the power function  $F(x) = x^{11}$  is not APN over  $\mathbb{F}_{2^4}$  while it satisfies the condition, that is, for every linearized polynomial  $L(x) = ax + bx^2 + cx^4 + dx^8$  over  $\mathbb{F}_{16}$ , the number of distinct images taken by the function  $x^{11} + L(x)$  is larger than or equal to 6.

## V. AN UPPER BOUND ON THE NONLINEARITY BY MEANS OF THE IMAGE SET SIZE

In [8] is also proved that the nonlinearity of any  $(n, m)$ -function  $F$  is bounded from above as follows:  $nl(F) \leq 2^{n-1} - \frac{2^{n+m-1}}{2^{m-1}} - 2^{n-1}$ . This bound is very weak, even if

we take for  $|Im(F)|$  the value which is the smallest and then the most in its favor, that is, according to Relation (6):  $|Im(F)| = \left\lceil \frac{2^{2n}}{(2^n-1)\delta_F+2^n} \right\rceil$ . Indeed, the bound says then that  $nl(F)$  is bounded above by approximately  $2^{n-1} - \frac{2^{m-n-1}(2^n-1)\delta_F+2^{m-1}-2^{n-1}}{2^{m-1}} \approx 2^{n-1} - \frac{1+\delta_F}{2} + 2^{n-m-1}$  and the bound is very far above the covering radius bound.

Let us show a much better bound with the same approach as for proving Proposition 1. We have seen that, thanks to the Cauchy-Schwarz inequality, we have  $|\{(x, y) \in \mathbb{F}_2^n; F(x) = F(y)\}| = \sum_{b \in Im(F)} |F^{-1}(b)|^2 \geq \frac{2^{2n}}{|Im(F)|}$ . We deduce  $\sum_{v \in \mathbb{F}_2^m} W_F^2(0_n, v) = \sum_{v \in \mathbb{F}_2^m; x, y \in \mathbb{F}_2^n} (-1)^{v \cdot (F(x)+F(y))} = 2^m |\{(x, y) \in \mathbb{F}_2^n; F(x) = F(y)\}| \geq \frac{2^{2n+m}}{|Im(F)|}$ . Hence, we have  $\sum_{v \in \mathbb{F}_2^m, v \neq 0_m} W_F^2(0_n, v) \geq \frac{2^{2n+m}}{|Im(F)|} - 2^{2n}$  and  $\max_{u \in \mathbb{F}_2^m, v \in \mathbb{F}_2^m, v \neq 0_m} W_F^2(u, v) \geq \frac{2^{2n+m}}{|Im(F)|} - 2^{2n}$  and  $\max_{v \in \mathbb{F}_2^m, v \neq 0_m} W_F^2(0_n, v) \geq \frac{2^{2n+m}}{|Im(F)|} - 2^{2n}$ . According to Relation (3), we deduce:

**Proposition 2:** For every positive integers  $n, m$  and every  $(n, m)$ -function, we have:

$$nl(F) \leq 2^{n-1} - \sqrt{\frac{2^{2n+m-2}}{|Im(F)|} - 2^{2n-2}}. \quad (9)$$

If we take again  $|Im(F)| = \left\lceil \frac{2^{2n}}{(2^n-1)\delta_F+2^n} \right\rceil$ , then  $nl(F)$  is bounded above by approximately  $2^{n-1} - \sqrt{\frac{2^{m-2}((2^n-1)\delta_F+2^n)-2^{2n-2}}{2^{m-1}}}$ , that is, if  $m = n$  for instance, by approximately  $2^{n-1} - \sqrt{2^{n-2}\delta_F}$ . This latter inequality is interesting when  $\delta_F > 2$  since it improves then upon the SCV bound. Note that, still for  $m = n$ , if  $\delta = 2$ , then

(9) writes  $nl(F) \leq 2^{n-1} - \sqrt{\frac{2^{3n-2}}{|Im(F)|} - 2^{2n-2}}$  and gives no information since, according to Relation (7), it is weaker than the SCV bound, except if  $|Im(F)| = \frac{2^{2n}}{3 \cdot 2^{n-2}}$  (in which case, the two bounds would coincide, but this number is not an integer). Let us now compare (9) for any  $m$  with the covering radius bound. We know from [17] that this latter bound is not tight for  $\frac{n}{2} < m$ . The bound (9) of Proposition 2 is strictly sharper than the covering radius bound if and only if we have

$\frac{2^{2n+m-2}}{|Im(F)|} - 2^{2n-2} > 2^{n-2}$ . We have then:

**Proposition 3:** For every positive integers  $n, m$  and every  $(n, m)$ -function, the bound (9) of Proposition 2 is sharper than the covering radius bound if and only if  $|Im(F)| < \frac{2^{n+m}}{2^{n+2m-1}}$ .

Note that when  $m$  ranges between  $\frac{n}{2}$  and  $n$ , this necessary and sufficient condition ranges from  $|Im(F)| \lesssim 2^m$  (i.e. no condition) to  $|Im(F)| \lesssim 2^{m-1}$ .

We see that the larger the image set size of  $F$ , the larger the upper bound of Proposition 3.

## VI. ON THE HAMMING DISTANCE BETWEEN DIFFERENTIALLY UNIFORM FUNCTIONS AND AFFINE VECTORIAL FUNCTIONS

Observing that, for every  $(n, m)$ -function  $G$ , we have  $d_H(F, G) = |\{x \in \mathbb{F}_2^n; F(x) \neq G(x)\}| \geq |Im(F+G)| - 1$ , since  $(F+G)(x)$  takes at least  $|Im(F+G)| - 1$  nonzero values, at least once each, we have then that the Hamming

distance from any differentially  $\delta$ -uniform  $(n, n)$ -function  $F$  to  $\mathcal{A}$  satisfies:

$$d_H(F, \mathcal{A}) \geq \left\lceil \frac{2^{2n}}{(2^n - 1)\delta + 2^n} \right\rceil - 1. \quad (10)$$

In particular, the Hamming distance between any APN function and  $\mathcal{A}$  is at least  $\left\lceil \frac{2^{2n}}{3 \cdot 2^n - 2} \right\rceil - 1$ .

The value of  $d_H(F, \mathcal{A})$ , contrary to the nonlinearity of  $F$ , is not directly linked to the linear attack, but as the nonlinearity does, it quantifies to which extent  $F$  is different from affine functions. This parameter has been studied in [6], [7], [15] where it was denoted in diverse ways (we shall keep here the notation  $d_H(F, \mathcal{A})$ ).

If for some  $x \in \mathbb{F}_2^n$ , some  $b, v \in \mathbb{F}_2^m$  and some linear  $(n, m)$ -function  $L$ , we have  $v \cdot F(x) \neq v \cdot (L(x) + b)$ , that is, denoting by  $L^*$  the adjoint operator of  $L$ , if we have  $v \cdot F(x) \neq L^*(v) \cdot x + v \cdot b$ , then we have  $F(x) \neq L(x) + b$ . Hence, denoting  $a = L^*(v)$  and  $\epsilon = v \cdot b$ , we have  $\{x \in \mathbb{F}_2^n; v \cdot F(x) \neq a \cdot x + \epsilon\} \subseteq \{x \in \mathbb{F}_2^n; F(x) \neq L(x) + b\}$ . This implies that  $d_H(F, \mathcal{A}) = \min_{b \in \mathbb{F}_2^m, L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) \neq L(x) + b\}| \geq \max_{v \in \mathbb{F}_2^m, v \neq 0_m} \min_{a \in \mathbb{F}_2^n, \epsilon \in \mathbb{F}_2} d_H(v \cdot F, a \cdot x + \epsilon) = \max_{v \in \mathbb{F}_2^m, v \neq 0_m} nl(v \cdot F) \geq \min_{v \in \mathbb{F}_2^m, v \neq 0_m} nl(v \cdot F) = nl(F)$ . The inequality  $d_H(F, \mathcal{A}) \geq nl(F)$  was already observed in [15] but we see with the inequalities above why these two parameters can be far from each other. Lower and upper bounds were given in [7] on  $d_H(F, \mathcal{A})$  when  $F$  is a bent function. Lower bounds on  $d_H(F, \mathcal{A})$  for a given function  $F$  do not imply lower bounds on the nonlinearity of  $F$ , but they give some insight on the chances that  $F$  can have good or bad nonlinearity.

Let us show now that a much stronger bound than (10) is valid:

**Proposition 4:** Let  $F$  be any  $\delta$ -uniform  $(n, m)$ -function, then we have:

$$d_H(F, \mathcal{A}) \geq 2^n - \sqrt{2^n + \delta(2^n - 1)}.$$

In particular, let  $F$  be any APN function, then we have:

$$d_H(F, \mathcal{A}) \geq 2^n - \sqrt{3 \cdot 2^n - 2}.$$

*Proof:* We have:

$$\begin{aligned} |F^{-1}(0_m)| &\leq \sqrt{\sum_{b \in \mathbb{F}_2^m} |F^{-1}(b)|^2} \\ &= \sqrt{|\{(x, y) \in (\mathbb{F}_2^n)^2; F(x) = F(y)\}|} \\ &= \sqrt{\sum_{a \in \mathbb{F}_2^n} |(D_a F)^{-1}(0_m)|} \\ &\leq \sqrt{2^n + \delta(2^n - 1)}. \end{aligned}$$

Applying this to  $F + L$  instead of  $F$ , we deduce:

$$d_H(F, \mathcal{A}) = |\{x \in \mathbb{F}_2^n; (F + L)(x) \neq 0_m\}| \geq 2^n - \sqrt{2^n + \delta(2^n - 1)}.$$

□

This bound shows that if  $\delta$  is small enough, function  $F$  contributes well as an S-box to the resistance against attacks by affine approximations.

## VII. ON THE MAXIMUM POSSIBLE VALUE OF $d_H(F, \mathcal{A})$

The number  $2^n - \sqrt{3 \cdot 2^n - 2}$  is rather close to  $2^n$  (which is of course an upper bound for  $d_H(F, \mathcal{A})$ ). This poses the question of determining what is the largest possible value of  $d_H(F, \mathcal{A})$  for all  $(n, m)$ -functions  $F$  (that is, finding for this other nonlinearity parameter, tight bounds similar to the covering radius bound for  $m < n$  and to the SCV bound for  $m \geq n$ , see e.g. [4]) and still more interestingly, what are the functions which reach it (which would be the equivalent of bent functions and of almost bent functions for this notion of nonlinearity). The following upper bound was given in a paper in Chinese and reproduced in [15]:  $d_H(F, \mathcal{A}) < (1 - 2^{-m})(2^n - 1)$ . The proof deals with character sums. Let us briefly present it (in a slightly simpler and more complete way): for every linear function  $L$ , we have  $|\{x \in \mathbb{F}_2^n; F(x) + L(x) = F(0_n)\}| = 2^{-m} \sum_{x \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{v \cdot (F(x) + L(x) + F(0_n))}$ . Denoting by  $\mathcal{L}$  the vector space of linear  $(n, m)$ -functions, we have that, if  $v \neq 0_m$ , then  $\sum_{L \in \mathcal{L}} (-1)^{v \cdot L(x)}$  equals  $|\mathcal{L}|$  if  $x = 0_n$  and equals 0 otherwise. This implies (distinguishing the cases  $v = 0_m$  and  $v \neq 0_m$ )  $\sum_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = F(0_n)\}| = (2^{n-m} + 2^{-m}(2^m - 1))|\mathcal{L}|$  and therefore:  $\max_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = F(0_n)\}| \geq 2^{n-m} + 1 - 2^{-m}$  and this gives indeed  $d_H(F, \mathcal{A}) \leq 2^n - 2^{n-m} - 1 + 2^{-m}$ . Note that since  $d_H(F, \mathcal{A})$  is an integer, this bound is equivalent to  $d_H(F, \mathcal{A}) \leq 2^n - 2^{n-m} - 1$  for  $m \leq n$  and to  $d_H(F, \mathcal{A}) \leq 2^n - 2$  for  $m \geq n$ .

In the next proposition, we obtain a bound that is slightly stronger when  $m < n$  (and is identical when  $m = n$ ).

**Proposition 5:** For every positive integers  $n, m$  and every  $(n, m)$ -function  $F$ , we have:

$$d_H(F, \mathcal{A}) \leq 2^n - \left\lceil 2^{\frac{n}{2}-m} \sqrt{2^n + 2^m - 1} \right\rceil,$$

where  $\mathcal{A}$  is the vector space of all affine functions over  $\mathbb{F}_2^n$  and  $d_H(F, \mathcal{A})$  is the minimum Hamming distance between  $F$  and affine functions.

*Proof:* For every linear  $(n, m)$ -function  $L$ , we have:

$$\begin{aligned} &\max_{b \in \mathbb{F}_2^m} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 \geq \\ &\frac{\sum_{b \in \mathbb{F}_2^m} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2}{2^m} = \\ &2^{-m} |\{(x, y) \in (\mathbb{F}_2^n)^2; F(x) + L(x) = F(y) + L(y)\}| = \\ &2^{-2m} \sum_{x, y \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{v \cdot (F(x) + F(y) + L(x) + L(y))}. \end{aligned}$$

We have, for every  $x, y \in \mathbb{F}_2^n$  and every nonzero  $v \in \mathbb{F}_2^m$  that  $\sum_{L \in \mathcal{L}} (-1)^{v \cdot L(x+y)}$  equals  $|\mathcal{L}|$  if  $x + y = 0_n$  and equals 0 otherwise. We deduce (distinguishing the cases  $v = 0_m$  and  $v \neq 0_m$ ):

$$\begin{aligned} &\sum_{L \in \mathcal{L}} \max_{b \in \mathbb{F}_2^m} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 \geq \\ &(2^{2n-2m} + (2^m - 1)2^{n-2m})|\mathcal{L}|, \end{aligned}$$

and therefore:

$$\begin{aligned} &\max_{L \in \mathcal{L}, b \in \mathbb{F}_2^m} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 \geq \\ &2^{2n-2m} + 2^{n-m} - 2^{n-2m} = \\ &2^{n-2m}(2^n + 2^m - 1). \end{aligned}$$

We deduce

$$\begin{aligned} d_H(F, \mathcal{A}) &= 2^n - \max_{L \in \mathcal{L}, b \in \mathbb{F}_2^m} |(F(x) + L(x) + b)^{-1}(0_m)| \\ &\leq 2^n - 2^{\frac{n}{2}-m} \sqrt{2^n + 2^m - 1}. \end{aligned}$$

This completes the proof.  $\square$

For  $m < n$ , we get  $d_H(F, \mathcal{A}) \leq 2^n - \left[ 2^{n-m} \sqrt{1 + 2^{m-n} - 2^{-n}} \right]$ , which is sharper than the bound  $d_H(F, \mathcal{A}) \leq 2^n - 2^{n-m} - 1$  of [15].

For  $m = n$ , we get  $d_H(F, \mathcal{A}) \leq 2^n - \left[ \sqrt{2 - 2^{-n}} \right] = 2^n - 2$ , the same as in [15].

For  $m > n$ , we get  $d_H(F, \mathcal{A}) \leq 2^n - \left[ 2^{\frac{n}{2}-\frac{m}{2}} \sqrt{1 + 2^{n-m} - 2^{-m}} \right]$  which may be worse by one unit than  $d_H(F, \mathcal{A}) \leq 2^n - 2$  proved in [15].

*Remark:* To avoid the loss of information due to the first inequality in the proof above, a slightly different approach consists in fixing  $b$  (taking later the best possible value), as done in [15], but keeping the square of  $|\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|$ . We have

$$\begin{aligned} &2^{2m} \sum_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 = \\ &\sum_{L \in \mathcal{L}} \sum_{\substack{x, y \in \mathbb{F}_2^n \\ v, w \in \mathbb{F}_2^m}} (-1)^{v \cdot (F(x) + L(x) + b) + w \cdot (F(y) + L(y) + b)} = \\ &\sum_{\substack{x, y \in \mathbb{F}_2^n \\ v, w \in \mathbb{F}_2^m}} (-1)^{v \cdot (F(x) + b) + w \cdot (F(y) + b)} \left( \sum_{L \in \mathcal{L}} (-1)^{v \cdot L(x) + w \cdot L(y)} \right). \end{aligned}$$

We have:

$$\sum_{L \in \mathcal{L}} (-1)^{v \cdot L(x) + w \cdot L(y)} = \begin{cases} |\mathcal{L}| & \text{if } x = y = 0_n \\ |\mathcal{L}| & \text{if } x = y \neq 0_n \text{ and } v = w \\ 0 & \text{if } x = y \neq 0_n \text{ and } v \neq w \\ |\mathcal{L}| & \text{if } x \neq y \text{ and } v = w = 0_m \\ 0 & \text{if } x \neq y \text{ and } v = w \neq 0_m \\ 0 & \text{if } x \neq y \text{ and } v \neq w. \end{cases}$$

This implies:

$$\begin{aligned} &\sum_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 = \\ &2^{-2m} \sum_{v, w \in \mathbb{F}_2^m} (-1)^{v \cdot (F(0_n) + b) + w \cdot (F(0_n) + b)} |\mathcal{L}| + \\ &2^{-2m} |\{(x, v) \in \mathbb{F}_2^n \times \mathbb{F}_2^m, x \neq 0_n\}| |\mathcal{L}| + \\ &2^{-2m} |\{(x, y) \in \mathbb{F}_2^n \times \mathbb{F}_2^n, x \neq y\}| |\mathcal{L}|. \end{aligned}$$

Here again, the value is maximal when  $b = F(0_n)$  and then we have

$$\begin{aligned} &\sum_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = F(0_n)\}|^2 = \\ &|\mathcal{L}| + 2^{-m} (2^n - 1) |\mathcal{L}| + 2^{-2m} 2^n (2^n - 1) |\mathcal{L}|. \end{aligned}$$

and therefore:

$$\begin{aligned} &\max_{L \in \mathcal{L}} |\{x \in \mathbb{F}_2^n; F(x) + L(x) = b\}|^2 \geq \\ &1 + 2^{-m} (2^n - 1) + 2^{-2m} 2^n (2^n - 1), \end{aligned}$$

that is:

$$d_H(F, \mathcal{A}) \leq 2^n - \left\lceil \sqrt{2^{2n-2m} + 2^{n-m} - 2^{n-2m} + 1 - 2^{-m}} \right\rceil.$$

This bound is probably (since we are taking the ceiling) exactly the same as the one in Proposition 5 (with maybe rare exceptions where it would be lower by 1).  $\diamond$

We can see with Proposition 5 and with the remark above that the bound of [15] is not easy to significantly improve with an approach by character sums. And for  $m = n$ , which is an important practical case, we have no improvement at all with such method. We shall now obtain, by a completely different approach, another bound which is stronger than both bounds for  $m \geq n$  (and more generally for  $m \geq n - \ln n$  where  $\ln$  is the natural logarithm).

Let us choose some  $a \in \mathbb{F}_2^n$  and  $n$  linearly independent elements  $a_1, \dots, a_n$  of  $\mathbb{F}_2^n$ ; there exists a unique affine  $(n, m)$ -function  $A$  such that  $A(a) = F(a)$  and  $A(a + a_i) = F(a + a_i)$  for  $i = 1, \dots, n$ . Let us briefly recall how this well-known fact can be shown: an  $(n, m)$ -function  $A$  is affine and such that  $A(a) = F(a)$  if and only if the function  $L(x) = F(a) + A(a + x)$  is linear (that is, affine and taking the zero value at  $0_n$ ), and thanks to the fact that  $(a_1, \dots, a_n)$  is a basis of the vector space  $\mathbb{F}_2^n$ , there exists a unique linear function  $L$  satisfying  $L(a_i) = F(a) + F(a + a_i)$  for  $i = 1, \dots, n$ . This writes  $A(a + a_i) = F(a + a_i)$ .

Then we have  $d_H(F, A) \leq 2^n - (n + 1)$  since  $A$  and  $F$  coincide at the  $n + 1$  distinct points  $a, a + a_1, \dots, a + a_n$ .

We have then:

*Proposition 6:* For every positive integer and every  $(n, m)$ -function  $F$ , we have:

$$d_H(F, \mathcal{A}) \leq 2^n - n - 1.$$

If this bound is tight, then, for every  $a \in \mathbb{F}_2^n$  and every linearly independent elements  $a_1, \dots, a_n$  of  $\mathbb{F}_2^n$ , any  $(n, m)$ -function  $F$  achieving it with equality must coincide with the affine function  $A$  defined above only at  $a, a + a_1, \dots, a + a_n$  (note that this condition is necessary but may not be sufficient). Hence, such  $F$  must satisfy  $F(a) + F(a + \sum_{i=1}^n \epsilon_i a_i) \neq L(\sum_{i=1}^n \epsilon_i a_i) = \sum_{i=1}^n \epsilon_i L(a_i) = \sum_{i=1}^n \epsilon_i (F(a) + F(a + a_i))$ , for every  $a \in \mathbb{F}_2^n$ , every basis  $(a_1, \dots, a_n)$  of  $\mathbb{F}_2^n$ , and every  $\epsilon \in \mathbb{F}_2^n$  of Hamming weight at least 2. If we look a little more precisely at the cases where  $w_H(\epsilon)$  is even and odd, respectively, we see that the condition is equivalent to saying that, for every even number  $2 \leq r \leq n$  of linearly independent elements  $a_1, \dots, a_r$ , the function  $F(x) + \sum_{i=1}^r F(x + a_i) + F(x + \sum_{i=1}^r a_i)$  never vanishes.

This is an interesting condition, which includes differential 2-uniformity (indeed, for  $r = 2$ , it is equivalent to saying that  $F$  is differentially 2-uniform). For  $n \geq 4$ , the condition seems much stronger than differential 2-uniformity. If we fix for instance  $r = 4$ , the resulting condition is equivalent to saying that  $D_{a_1} D_{a_2} F(x) + D_{a_3} D_{a_4} F(x) + D_{a_1+a_2} D_{a_3+a_4} F(x)$  never vanishes; hence, not only each of these three second-order derivatives would not vanish; their sum would not either. However, we have:

*Proposition 7:* Let  $n$  and  $m$  be any positive integers and  $F$  be any differentially 2-uniform  $(n, m)$ -function. Then  $F$  satisfies that, for every linearly independent elements  $a_1, \dots, a_4$  of



$\mathbb{F}_2^n$ , the function  $F(x) + \sum_{i=1}^4 F(x + a_i) + F(x + \sum_{i=1}^4 a_i)$  never vanishes if and only if:

$$\sum_{u,v \in \mathbb{F}_2^n, v \neq 0_n} W_F^6(u, v) = -2^{6n} + 18 \cdot 2^{4n+m} - 39 \cdot 2^{3n+m} + 22 \cdot 2^{2n+m}.$$

No such  $(n, m)$ -function  $F$  exists if  $m \leq 2n - 5$  or  $m \leq n = 4$ .

*Proof:* For obtaining such characterization, we shall need to address all cases for  $a_1, \dots, a_4$ , whether they are linearly independent or not. Let us then first study the behavior of the function  $\phi_{a_1, a_2, a_3, a_4}(x) := F(x) + \sum_{i=1}^4 F(x + a_i) + F(x + \sum_{i=1}^4 a_i)$  when  $a_1, \dots, a_4$  are linearly dependent.

If one element among  $a_1, \dots, a_4$  is equal to zero (say  $a_4 = 0_n$ ), then the function  $\phi_{a_1, a_2, a_3, a_4}(x)$  equals  $F(x + a_1) + F(x + a_2) + F(x + a_3) + F(x + a_1 + a_2 + a_3) = D_{a_1+a_2} D_{a_1+a_3} F(x + a_1)$  and since  $F$  is differentially 2-uniform:

- either  $a_1 + a_2$  and  $a_1 + a_3$  are linearly dependent (that is,  $a_1 = a_2$  or  $a_1 = a_3$  or  $a_2 = a_3$ ) and  $\phi_{a_1, a_2, a_3, 0_n}(x)$  is the zero function,

- or they are linearly independent (that is,  $a_1, a_2, a_3$  are distinct) and  $\phi_{a_1, a_2, a_3, 0_n}(x)$  does not vanish.

If no element is zero among  $a_1, \dots, a_4$  and two elements are equal (say  $a_1 = a_2$ ), then  $\phi_{a_1, a_2, a_3, a_4}(x)$  equals  $D_{a_3} D_{a_4} F(x)$  and since  $F$  is differentially 2-uniform:

- either  $a_3$  and  $a_4$  are linearly dependent (that is,  $a_3 = a_4$ ) and  $\phi_{a_1, a_2, a_3, a_4}(x)$  is the zero function,

- or they are linearly independent (that is, distinct) and  $\phi_{a_1, a_2, a_3, a_4}(x)$  does not vanish.

If no sum of at least one and at most two elements among  $a_1, \dots, a_4$  is zero and the sum of three elements is zero (say  $a_2 + a_3 + a_4 = 0_n$ ), then  $\phi_{a_1, a_2, a_3, a_4}(x)$  equals  $D_{a_3} D_{a_4} F(x)$  and we are back to the same situation, but then  $a_3$  and  $a_4$  cannot be linearly dependent since  $a_3 = a_4$  would imply  $a_2 = 0_n$  and then  $\phi_{a_1, a_2, a_3, a_4}(x)$  does not vanish.

If no sum of at least one and at most three elements among  $a_1, \dots, a_4$  is zero and the sum of all four elements is zero, then  $\phi_{a_1, a_2, a_3, a_4}(x)$  equals  $F(x + a_1) + F(x + a_2) + F(x + a_3) + F(x + a_4) = D_{a_1+a_2} D_{a_1+a_3} F(x + a_1)$  and  $a_1 + a_2$  and  $a_1 + a_3$  cannot be linearly dependent since this would mean that  $a_1 = a_2$  or  $a_1 = a_3$  or  $a_2 = a_3$ , which is excluded; then  $\phi_{a_1, a_2, a_3, a_4}(x)$  does not vanish since  $F$  is differentially 2-uniform.

Summarizing, the condition in Proposition 7 is equivalent to: for every  $a_1, \dots, a_4$  such that:

- one element is zero and the others are not distinct, or two elements are equal and non-zero and the two others are equal and nonzero too, then  $\phi_{a_1, a_2, a_3, a_4}(x)$  is the zero function,
- in all the other cases,  $\phi_{a_1, a_2, a_3, a_4}(x)$  does not vanish.

The number  $N$  of quadruples  $(a_1, a_2, a_3, a_4)$  such that one element is zero and the others are not distinct, or two elements are equal and the two others are equal too, can be evaluated as follows. Counting each case once and once only, by considering successively the subcases where the number of zero elements equals 4, 3, 2, 1, 0 gives:  $N = 1 + 4(2^n - 1) + 6(2^n - 1) + 4(2^n - 1)(3 \cdot 2^n - 5) + (2^n - 1)(6 \cdot 2^n - 11) = 1 + (2^n - 1)(18 \cdot 2^n - 21)$ . Hence we have:  $N = 18 \cdot 2^{2n} - 39 \cdot 2^n + 22$ .

Since, for every element  $b$  of  $\mathbb{F}_2^m$ , the sum  $\sum_{v \in \mathbb{F}_2^m} (-1)^{v \cdot b}$  equals  $2^m$  if  $b = 0_m$  and equals zero otherwise, the condition in Proposition 7 is equivalent to:

$$\sum_{\substack{x, a_1, \dots, a_4 \in \mathbb{F}_2^n \\ v \in \mathbb{F}_2^m}} (-1)^{v \cdot (F(x) + \sum_{i=1}^4 F(x + a_i) + F(x + \sum_{i=1}^4 a_i))} = 2^{n+m} N.$$

Using the inverse Walsh transform relation, we have:

$$\begin{aligned} 2^{6n} \sum_{\substack{x, a_1, \dots, a_4 \in \mathbb{F}_2^n \\ v \in \mathbb{F}_2^m}} (-1)^{v \cdot (F(x) + \sum_{i=1}^4 F(x + a_i) + F(x + \sum_{i=1}^4 a_i))} &= \\ \sum_{\substack{x, a_1, \dots, a_4, u_1, \dots, u_6 \in \mathbb{F}_2^n \\ v \in \mathbb{F}_2^m}} \prod_{i=1}^6 W_F(u_i, v) (-1)^{(u_1 + \dots + u_6) \cdot x + \sum_{i=1}^4 (u_{i+1} + u_6) \cdot a_i} &= \\ = 2^{5n} \sum_{u, v \in \mathbb{F}_2^n} W_F^6(u, v). \end{aligned}$$

Hence, the condition is equivalent to:

$$\sum_{u, v \in \mathbb{F}_2^n} W_F^6(u, v) = 18 \cdot 2^{4n+m} - 39 \cdot 2^{3n+m} + 22 \cdot 2^{2n+m},$$

that is to:

$$\sum_{u, v \in \mathbb{F}_2^n, v \neq 0_n} W_F^6(u, v) = -2^{6n} + 18 \cdot 2^{4n+m} - 39 \cdot 2^{3n+m} + 22 \cdot 2^{2n+m}.$$

This is impossible for  $m \leq 2n - 5$ , since the number on the right-hand side is then smaller than or equal to  $-\frac{7}{16} \cdot 2^{6n} - \frac{39}{32} \cdot 2^{5n} + \frac{11}{16} \cdot 2^{4n}$  and is therefore negative. It is also negative if  $m \leq n = 4$ .  $\square$

Hence, for every  $(n, m)$  such that  $m \leq 2n - 5$  or  $m \leq n = 4$ , the inequality in Proposition 6 is in fact strict.

*Remark:* Proposition 7 proves that, if  $m \leq 2n - 5$  or  $m \leq n = 4$ , then for every  $(n, m)$ -function  $F$ , there exist a basis  $(a_1, \dots, a_n)$  of  $\mathbb{F}_2^n$  and two vectors  $x, \epsilon$  in  $\mathbb{F}_2^n$ , such that  $w_H(\epsilon) \geq 2$  and  $F(x) + F(x + \sum_{i=1}^n \epsilon_i a_i) + \sum_{i=1}^n \epsilon_i (F(x) + F(x + a_i)) = 0_m$ . It would be interesting to determine whether, for every basis  $(a_1, \dots, a_n)$  of  $\mathbb{F}_2^n$ , there exist two vectors  $x, \epsilon$  in  $\mathbb{F}_2^n$  having such property, but it seems difficult to do so. Denoting by  $(e_1, \dots, e_n)$  the canonical basis of  $\mathbb{F}_2^n$  (of those Hamming weight 1 vectors), this is equivalent (by composing  $F$  by the linear automorphism mapping  $(a_1, \dots, a_n)$  to  $(e_1, \dots, e_n)$ ), to saying that, for every  $(n, m)$ -function  $F$ , there exist two vectors  $x, \epsilon$  in  $\mathbb{F}_2^n$  such that  $w_H(\epsilon) \geq 2$  and  $D_\epsilon F(x) + \sum_{i=1}^n \epsilon_i D_{e_i} F(x) = 0_m$ . It seems difficult to check if there can exist  $F$  such that, for every  $\epsilon$  in  $\mathbb{F}_2^n$  of Hamming weight at least 2 and every  $x$ , this latter expression is nonzero. We recall that we have seen that the case where  $w_H(\epsilon)$  is odd reduces itself to the case where  $w_H(\epsilon)$  is even, so we shall assume that we are in this latter case. If we use the inverse Walsh transform relation again, the number of  $x$  such that  $D_\epsilon F(x) + \sum_{i=1}^n \epsilon_i D_{e_i} F(x) = 0_m$  equals, denoting the support of  $\epsilon$  by  $I$  (whose size is even) and writing the elements



of  $(\mathbb{F}_2^n)^I$  in the form  $U = (u_i)_{i \in I}$ :

$$\begin{aligned} & 2^{-m} \sum_{x \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{v \cdot (F(x) + F(x+\epsilon) + \sum_{i \in I} F(x+e_i))} = \\ & 2^{-(|I|+2)n-m} \sum_{\substack{x, u_\emptyset, u_\epsilon \in \mathbb{F}_2^n \\ U \in (\mathbb{F}_2^n)^I, v \in \mathbb{F}_2^m}} W_F(u_\emptyset, v) W_F(u_\epsilon, v) \prod_{i \in I} W_F(u_i, v) \\ & \quad \cdot (-1)^{(u_\emptyset + u_\epsilon + \sum_{i \in I} u_i) \cdot x + \sum_{i \in I} (u_\epsilon + u_i) \cdot e_i} = \\ & 2^{-(|I|+1)n-m} \sum_{\substack{u_\epsilon \in \mathbb{F}_2^n \\ U \in (\mathbb{F}_2^n)^I, v \in \mathbb{F}_2^m}} W_F(u_\epsilon + \sum_{i \in I} u_i, v) W_F(u_\epsilon, v) \\ & \quad \cdot \prod_{i \in I} W_F(u_i, v) (-1)^{\sum_{i \in I} (u_\epsilon + u_i) \cdot e_i}. \end{aligned}$$

It seems difficult to go further.  $\diamond$

Reference [15] conjectures that  $d_H(F, \mathcal{A}) \leq (1-2^{-m})(2^n - 2^{\frac{n}{2}})$ . We see that, according to Proposition 4, APN functions are good candidates for approaching this conjectured bound (if it is true) or for disproving it (if it is false).

### VIII. AN UPPER BOUND ON THE NONLINEARITY BY MEANS OF THE MINIMUM DISTANCE TO AFFINE FUNCTIONS

We have recalled in Section VI that, for any  $(n, m)$ -function  $F$ , we have  $nl(F) \leq d_H(F, \mathcal{A})$ . Proposition 2 implies another upper bound on the nonlinearity by an expression depending on  $d_H(F, \mathcal{A})$ . Indeed, let us apply this proposition to  $F + A$  where  $A$  is the best affine approximation of  $F$ . Since  $F + A$  equals 0 at  $2^n - d_H(F, \mathcal{A})$  points, we have  $|Im(F + A)| \leq d_H(F, \mathcal{A}) + 1 = d_H(F, \mathcal{A}) + 1$ . Hence:

*Corollary 2:* For every positive integers  $n, m$  and every  $(n, m)$ -function  $F$ , we have:

$$nl(F) \leq 2^{n-1} - \sqrt{\frac{2^{2n+m-2} - 2^{2n-2}}{\frac{d_H(F, \mathcal{A})+1}{2^m-1} - 1}}.$$

This bound must be compared with the bound  $nl(F) \leq d_H(F, \mathcal{A})$ . It improves upon it if and only if  $2^{n-1} - \sqrt{\frac{2^{2n+m-2} - 2^{2n-2}}{\frac{d_H(F, \mathcal{A})+1}{2^m-1} - 1}} < d_H(F, \mathcal{A})$ , that is,  $\frac{2^{2n+m-2}}{d_H(F, \mathcal{A})+1} > 2^{2n-2} + (2^m - 1)(2^{n-1} - d_H(F, \mathcal{A}))^2$ , that is,  $(2^m - 1)(d_H(F, \mathcal{A}))^3 + (2^m - 1)(1 - 2^n)(d_H(F, \mathcal{A}))^2 + (2^{2n-2} + (2^m - 1)(2^{2n-2} - 2^n))d_H(F, \mathcal{A}) + 2^{2n-2} + (2^m - 1)2^{2n-2} - 2^{2n+m-2} < 0$ . This inequality has the form  $A(d_H(F, \mathcal{A}))^3 + B(d_H(F, \mathcal{A}))^2 + C d_H(F, \mathcal{A}) + D < 0$  with  $A = 2^m - 1 > 0$ ,  $B = (2^m - 1)(1 - 2^n) < 0$ ,  $C = 2^{2n+m-2} - (2^m - 1)2^n > 0$  and  $D = 0$ .

We have then  $2^{n-1} - \sqrt{\frac{2^{2n+m-2} - 2^{2n-2}}{\frac{d_H(F, \mathcal{A})+1}{2^m-1} - 1}} < d_H(F, \mathcal{A})$  when  $d_H(F, \mathcal{A})$  is between the two zeros  $\frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \frac{(2^m-1)(2^n-1) \pm \sqrt{(2^m-1)\sqrt{2^{2n+m+1}+2^m-2^{2n}-2^{2n+1}-1}}}{2(2^m-1)}$ , which are located between 0 and  $2^n$ .

### IX. AN IMPROVEMENT OF THE LOWER BOUND OF PROPOSITION 1

Let us show that the bound of Proposition 1 can be made stronger for some functions. Let us denote by  $\Delta$  the set  $\{x + y; (x, y) \in (\mathbb{F}_2^n)^2, x \neq y, F(x) = F(y)\}$ . For every

nonzero  $a \notin \Delta$ , we have  $|(D_a F)^{-1}(0_m)| = 0$ . Let us assume that  $F$  is not injective. Then we have  $|\Delta| > 0$ . We can then refine the calculations that led to Proposition 1 as follows:  $\sum_{a \in \Delta} |(D_a F)^{-1}(0_m)| = \sum_{a \in \mathbb{F}_2^n} |(D_a F)^{-1}(0_m)| - 2^n \geq$

$\frac{2^{2n}}{|Im(F)|} - 2^n$ , and we deduce that there exists  $a \in \mathbb{F}_2^n$ , nonzero, such that  $|D_a F^{-1}(0_m)| \geq \frac{2^{2n}}{\frac{|Im(F)|}{|\Delta|} - 2^n}$ . Hence:

*Proposition 8:* For every non-injective  $(n, m)$ -function, the differential uniformity of  $F$  satisfies:

$$\delta_F \geq \left\lceil \frac{\frac{2^{2n}}{\frac{|Im(F)|}{|\Delta|} - 2^n}}{|\Delta|} \right\rceil, \quad (11)$$

where  $\Delta = \{x + y; (x, y) \in (\mathbb{F}_2^n)^2, x \neq y, F(x) = F(y)\}$ .

Relation (11) improves upon Proposition 1 when  $|\Delta| < 2^n - 1$ .

*Remark:* We have:

$$\begin{aligned} |\Delta| & \leq \frac{1}{2} |\{(x, y) \in (\mathbb{F}_2^n)^2; F(x) = F(y)\}| - 2^{n-1} \\ & = 2^{-(m+1)} \sum_{x, y \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{v \cdot (F(x) + F(y))} - 2^{n-1} \\ & = 2^{-(m+1)} \sum_{v \in \mathbb{F}_2^m} W_F^2(0_n, v) - 2^{n-1}, \end{aligned}$$

and this bound is tight since it is achieved by those functions such that, in the multiset  $\{x + y; (x, y) \in (\mathbb{F}_2^n)^2, x \neq y, F(x) = F(y)\}$ , each value is matched at most twice.  $\diamond$

### X. CONCLUSION

In this paper, we have revisited and clarified a four year old result on the size of the image set of any differentially uniform function, and in particular of any APN function, and we have studied its consequences. We have derived an upper bound on the nonlinearity of vectorial functions by means of their image set size. We have also shown that differentially uniform functions lie at large Hamming distance from affine functions and preserve then the block ciphers in which they are used as substitution boxes from attacks based on affine approximation as well. The fact that the image set size of the sum of any differentially uniform function with any linear function is bounded from below may provide a new theoretical and computational approach of differentially uniform functions, and in particular of APN functions, which is worth future studies. There is also a need to study the Hamming distance between vectorial functions (possibly APN) and affine functions, for which tight bounds similar to the covering radius bound and to the Sidelnikov-Chabaud-Vaudenay bound are missing, as well as the knowledge of functions similar to bent functions for this distance.

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